

Milwaukee Housing Data Exploration

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In this exploration, we want to see if we can predict Milwaukee home prices using housing-related variables. Given the number of variables, we will use multidimensional scaling (MDS) to represent the data in two dimensions. The data is sourced from Milwaukee Open Data, which can be found here:

<https://data.milwaukee.gov/dataset/property-sales-data>

This exploration is crucial for understanding what impacts house prices in Milwaukee, which is particularly useful for first-time home buyers. By uncovering these relationships, homeowners are better able to understand housing options that are affordable.

```
Data = read.csv("/Users/elliottbarta/Downloads/housing2024.csv")
Data = drop_na(Data)
Data_clean = Data[!apply(Data == "NULL", 1, any), ]
Data_clean$PropType = factor(Data_clean$PropType)
Data_clean$CondoProject = factor(Data_clean$CondoProject)
Data_clean$Style = factor(Data_clean$Style)
Data_clean$Extwall = factor(Data_clean$Extwall)

Data_clean = as.data.frame(lapply(Data_clean, function(x) {
  if(is.factor(x)) {
    as.numeric(x)
  } else if(is.character(x)) {
    as.numeric(as.character(x))
  } else {
    x
  }
}))
```

```
## Warning in FUN(X[[i]], ...): NAs introduced by coercion
```

```
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```

Here, we clean the data to remove any rows with empty cells. We also convert categorical variables into factors for analysis.

```
DataFilt <- Data_clean %>% select(-Address, -Sale_date, -taxkey, -PropertyID, -Sale_price)
summary(DataFilt)
```

```
##      PropType      CondoProject      District      nbhd
## Min.   :1.000   Min.    : 1.00   Min.    : 1.00   Min.    : 40
## 1st Qu.:4.000   1st Qu.: 1.00   1st Qu.: 4.00   1st Qu.:1740
## Median :4.000   Median : 1.00   Median : 7.00   Median :2900
## Mean   :3.695   Mean    :13.85   Mean    : 7.77   Mean    :3132
## 3rd Qu.:4.000   3rd Qu.: 1.00   3rd Qu.:11.00   3rd Qu.:4520
## Max.   :4.000   Max.    :175.00   Max.    :15.00   Max.    :6460
##
##      Style      Extwall      Stories      Year_Built      Rooms
```

```
## Min. : 1.00 Min. : 1.00 Min. :1.0 Min. :1853 Min. : 0.000
## 1st Qu.:11.00 1st Qu.: 3.00 1st Qu.:1.0 1st Qu.:1921 1st Qu.: 5.000
## Median :19.00 Median : 3.00 Median :1.0 Median :1947 Median : 6.000
## Mean :16.81 Mean : 4.63 Mean :1.4 Mean :1941 Mean : 6.921
## 3rd Qu.:24.00 3rd Qu.: 6.00 3rd Qu.:2.0 3rd Qu.:1957 3rd Qu.: 9.000
## Max. :40.00 Max. :16.00 Max. :5.0 Max. :2020 Max. :70.000
##
## NA's :6
## FinishedSqft Units Bdrms Fbath
## Min. : 325 Min. : 1.000 Min. : 0.000 Min. :0.000
## 1st Qu.: 1065 1st Qu.: 1.000 1st Qu.: 3.000 1st Qu.:1.000
## Median : 1357 Median : 1.000 Median : 3.000 Median :2.000
## Mean : 1557 Mean : 1.316 Mean : 3.491 Mean :1.554
## 3rd Qu.: 1884 3rd Qu.: 2.000 3rd Qu.: 4.000 3rd Qu.:2.000
## Max. :32000 Max. :37.000 Max. :43.000 Max. :6.000
##
## Hbath Lotsize
## Min. :0.0000 Min. : 0
## 1st Qu.:0.0000 1st Qu.: 3660
## Median :0.0000 Median : 4998
## Mean :0.3179 Mean : 5121
## 3rd Qu.:1.0000 3rd Qu.: 6377
## Max. :3.0000 Max. :79279
##
```

We remove Address, Sale_date, taxkey, and PropertyID, since they are unique identifiers and have no predictive utility.

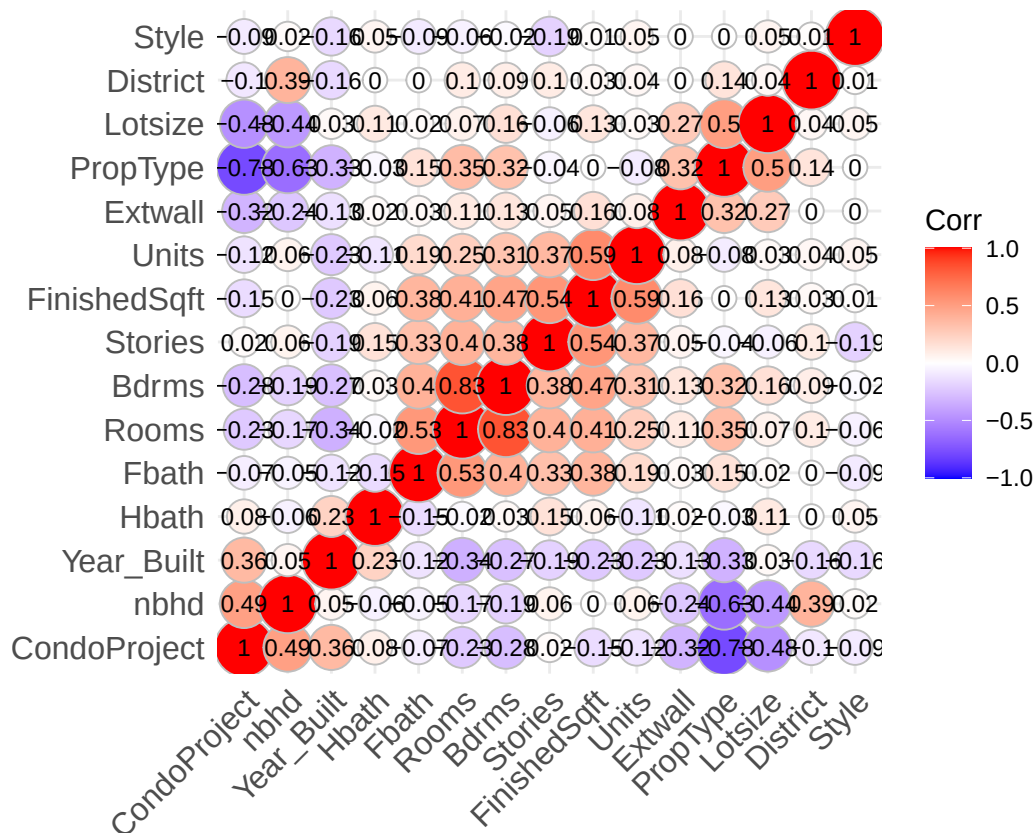
The following are the meaningful variables used in this analysis:

Style: Style of the house (e.g., Colonial, Ranch, Condo Townhouse). District: District where the house is located. Lotsize: Lot size of the property. PropType: Property type (e.g., Commercial, Residential, Condominium). Extwall: Exterior wall material. Units: Number of units in the house. FinishedSqft: Finished square footage of the property. Stories: Number of stories. Bdrms: Number of bedrooms. Rooms: Number of rooms. Fbath: Number of full bathrooms. Hbath: Number of half bathrooms. Year_Built: Year the property was built. CondoProject: Condo project the property is listed under.

```
cm = cor(DataFilt, use = "pairwise.complete.obs")
anyNA(cm)
```

```
## [1] FALSE
```

```
ggcorrplot(cm, hc.order = TRUE, lab = TRUE,
           lab_size = 3, method = "circle",
           colors = c("blue", "white", "red"))
```



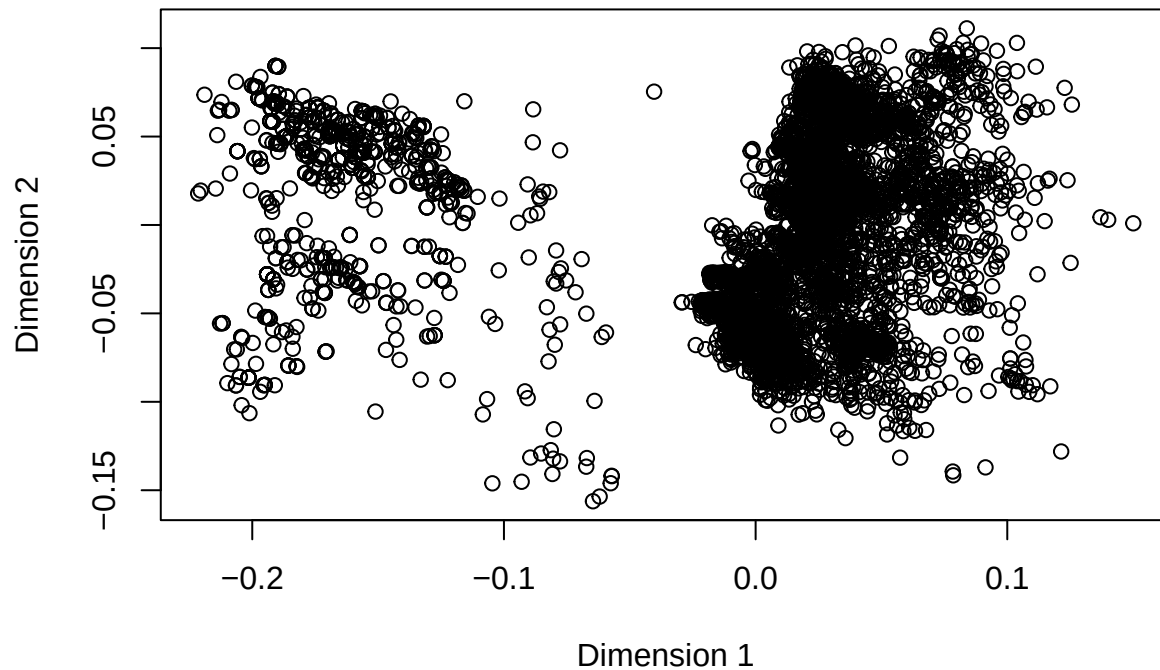
Next, we will create a correlation heat map to see how each variable correlates with each other. From the map, we can see that no variables are strongly correlated with one another other than between property type and condo project.

```
dist = daisy(DataFilt, metric = "gower" )
mds_result = cmdscale(dist, k =2)
```

To handle both qualitative and categorical data, we use the Gower metric. This produces a distance matrix that can be used in multidimensional scaling. MDS allows us to visualize variable relationships while also reducing the number of variables before clustering or regression.

```
plot(mds_result[,1], mds_result[,2],
     xlab = "Dimension 1", ylab = "Dimension 2",
     main = "Classical MDS on Gower Distance")
text(mds_result[,1], mds_result[,2], labels = rownames(df), pos = 3)
```

Classical MDS on Gower Distance



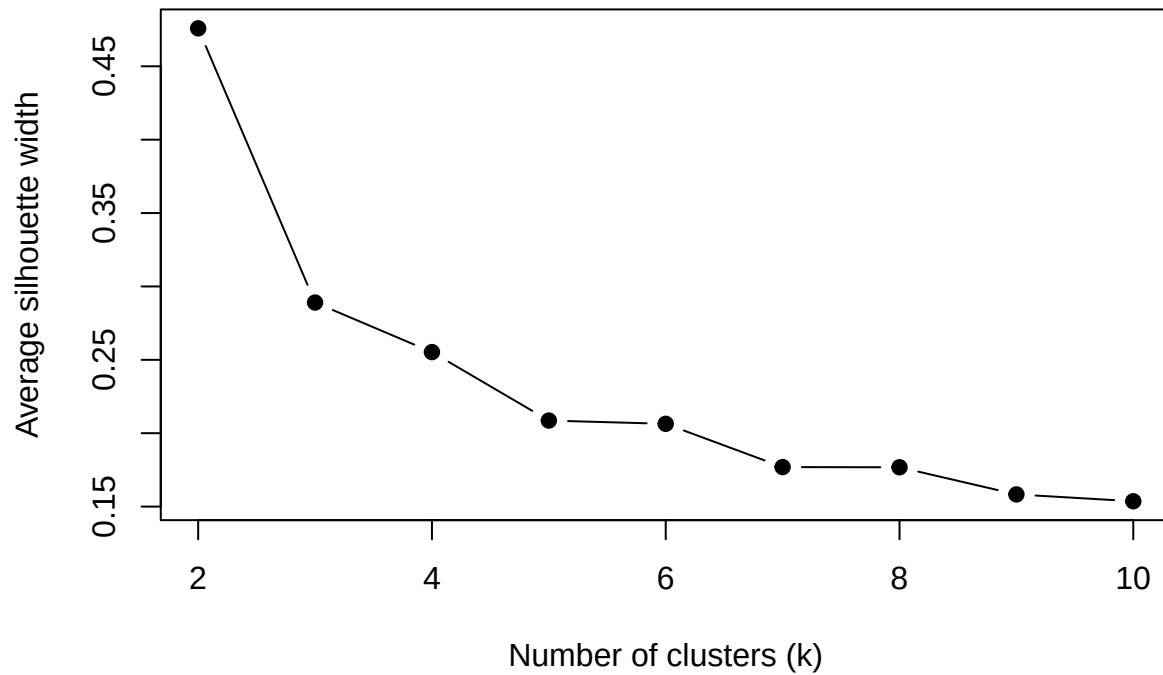
After applying MDS, we see that most houses cluster on the right side of the plot. The left side is sparse, with a faint distinction between the upper and lower halves of the graph.

```
sil_widths = numeric(10)

for (k in 2:10) {
  km = kmeans(mds_result, centers = k)
  ss = silhouette(km$cluster, dist)
  sil_widths[k] = mean(ss[, 3])
}

plot(2:10, sil_widths[2:10], type = "b", pch = 19,
     xlab = "Number of clusters (k)", ylab = "Average silhouette width",
     main = "Silhouette Plot of Housing Clusters")
```

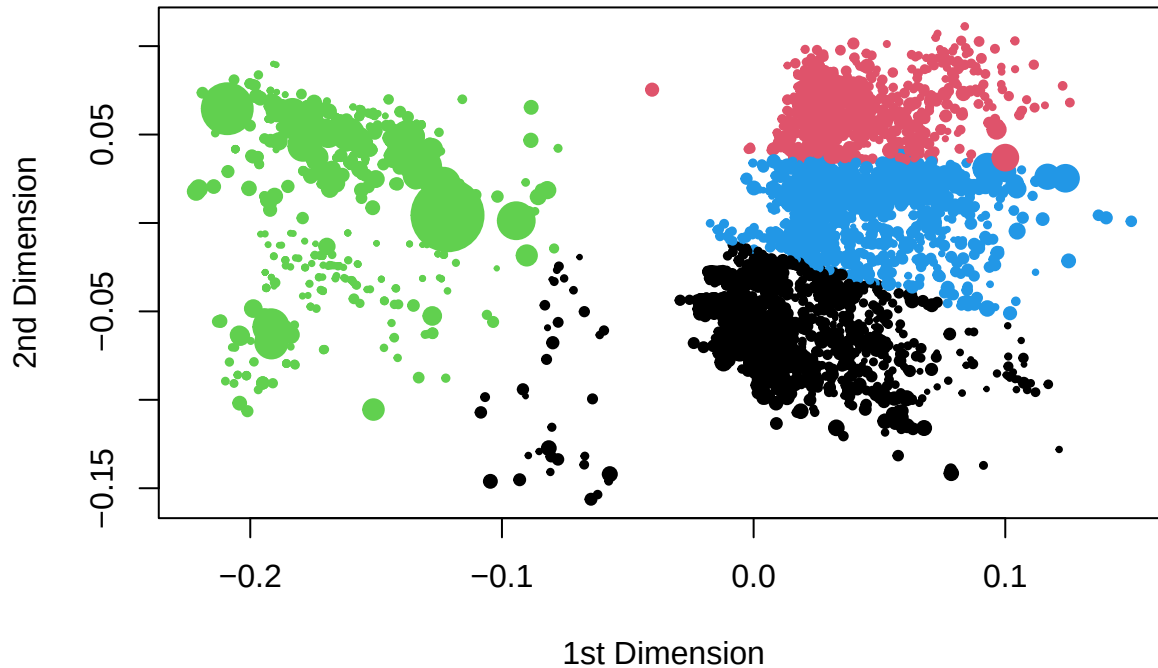
Silhouette Plot of Housing Clusters



Using a silhouette plot, we can see that having four clusters is appropriate. This is because of the semi-prominent bend seen at $k = 4$.

```
clust = kmeans(mds_result, iter.max = 100, nstart = 10, centers = 4)
plot(mds_result,
     col = clust$cluster,
     pch = 19,
     cex = scales::rescale(Data_clean$Sale_price, to = c(0.25, 5)),
     main = "MDS and K-Means (Point Size = Sale Price)",
     xlab = "1st Dimension",
     ylab = "2nd Dimension")
```

MDS and K-Means (Point Size = Sale Price)



With four clusters, the data points clearly split into groups. Interestingly, the houses with the highest sale prices are located on the left side of the graph. We can now examine correlations between the original variables and the reduced dimensions.

```
cors1 = sapply(DataFilt, function(v) cor(as.numeric(v), mds_result[,1]))
cors2 = sapply(DataFilt, function(v) cor(as.numeric(v), mds_result[,2]))

cors_table = data.frame(Dim1 = cors1, Dim2 = cors2)
```

cors_table

##	Dim1	Dim2
## PropType	0.90333360	-0.076799789
## CondoProject	-0.85046338	0.087878099
## District	0.02610870	-0.928474820
## nbhd	-0.71835053	-0.510774822
## Style	0.05554476	-0.004210484
## Extwall	0.54348152	-0.006958976
## Stories	NA	NA
## Year_Built	-0.40698460	0.337389272
## Rooms	0.38980686	-0.210933745
## FinishedSqft	0.18993514	-0.165496637
## Units	0.09501263	-0.172095896
## Bdrms	0.40585654	-0.191983687
## Fbath	0.18437883	-0.114042743
## Hbath	-0.02505596	0.060927193
## Lotsize	0.53968325	0.049483180

From the correlation table, property type, condo project, and neighborhood are highly correlated with the first dimension, while district is strongly related to the second dimension. Other variables show moderate importance.

```
model = lm(Data_clean$Sale_price ~ mds_result )
summary(model)
```

```
##
## Call:
## lm(formula = Data_clean$Sale_price ~ mds_result)
##
## Residuals:
```

	Min	1Q	Median	3Q	Max
	-263502	-80163	-12393	48314	3340633

```
##
## Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	237155	2082	113.934	< 2e-16 ***
mds_result1	-291543	28958	-10.068	< 2e-16 ***
mds_result2	-170668	40124	-4.254	2.14e-05 ***

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 152400 on 5361 degrees of freedom
## Multiple R-squared:  0.0218, Adjusted R-squared:  0.02143
## F-statistic: 59.73 on 2 and 5361 DF,  p-value: < 2.2e-16
```

We will then use the dimensions from MDS as variables in a linear regression model. We want to see whether they are significant in predicting housing prices. From the summary, we can see that both dimensions are significant at a 95% level of confidence. An increase in both the first and second dimension would lead to a lower house price. When compared to the cluster graph, this is somewhat surprising, as although the larger dots have negative values for the first dimension, they also have positive values for the second dimension. Moreover, while R^2 is not overall critical in establishing the efficacy of the model, a value of 0.0218 may show the limitations of MDS and the necessity of more independent variables.

Unsurprisingly, new homeowners should focus on location as a key indicator as to why certain property prices are larger than others.